

Congestion-EQ: Total-Variation Contracts for Queue/RTT Leakage in Anonymous Overlays and Anonymous DHT Lookups

Draft v0.31 (2026-02-28) (archive v125)

Abstract

Anonymous overlays and anonymous DHT lookup compilers typically treat *destination/contact traces* as the dominant metadata channel, and then equalize or hide those traces via scheduling, padding, and randomized delegation. This paper isolates a distinct leakage tier: *congestion observables* (queueing delay, RTT, loss/ECN marks, and their time series), which can amplify tiny secret-dependent biases across repeated operations. We propose *Congestion-EQ*, a history-conditioned total-variation (TV) contract that treats congestion as an explicit channel in the compiler calculus. We derive slack-sensitivity laws in the M/M/1 setting, give a compositional contract with an accountant for repeats/resets/public randomness, and connect the contract to an Anonymous-DHT microstudy. Additional robustness microstudies (heavy-tailed service and bursty/correlated arrivals, including a DHT-like fanout+retry burst model) are summarized in Section 10; full plots and scripts are available on request.

Scope. We treat congestion observables (queueing delay/RTT/loss time series) as a first-class metadata channel and define a history-conditioned TV contract (Congestion-EQ) suitable for repeated-use accounting.

Imports. We import the series' lookup-compiler vocabulary and the notion of TV certificates for contact-trace equalization.[1, 2] Stop-time padding and schedule compilation are treated as orthogonal components; Congestion-EQ is a *separate tier* that can remain informative even when contact traces are equalized. Notation conventions for keys, state, transcripts, and transcript projections follow the series notation sheet (Synthesis 8).[47]

Exports. This paper exports the Congestion-EQ contract definition, analytic slack-sensitivity laws in the M/M/1 setting, and compositional rules (stages, epochs, resets) designed to bind into certificate/log workflows developed in Paper III.[5]

Artifact policy. This archive is source-only; supporting plots and scripts are available on request.[46]

1 Introduction

Prior work in this series develops total-variation (TV) contracts and certificates for metadata-hiding overlay lookups, focusing on destination/contact traces and termination-time leakage [1, 2]. This paper adds a distinct leakage tier: *congestion observables* (queueing delay, RTT, loss/ECN marks, and their time series) that can remain informative even when contact traces are equalized.

Series map. Paper I (this paper) defines the Congestion-EQ TV contract and an explicit accountant for repetition and reset. Paper II, PSC-Q [4], gives public-service-calendar and substitution-padding mechanisms that help meet Congestion-EQ budgets without driving utilization to saturation. Paper III, W-Congestion-EQ [5], shows how to publish certificate-ready knob choices and anchor them in transparency logs.

Why congestion is different. Congestion is nonlinear in load, stateful, and naturally observed. Congestion/latency side channels have long been studied in Tor and related systems, from early circuit-clogging latency probing [30, 33, 32] to congestion-control and padding interactions [35, 36]; fully remote latency side channels [29, 31] and RTT-aware trace transduction [45] further widen the attacker surface. Congestion-control dynamics are themselves measurable: recent tools infer deployed congestion-control algorithms from queue/RTT traces at scale [37, 38], and custom congestion control in circumvention settings can be fingerprintable [39]. Complementary defenses use principled traffic shaping: NetShaper provides differential-privacy (DP) guarantees while adapting shaping decisions to workload and congestion conditions [40]; Pacer provides an end-to-end systems mitigation for packet-timing side channels that explicitly respects flow control, congestion control, and loss recovery [41]. As slack shrinks, small secret-dependent load biases can cause large shifts in delay/loss distributions (queue timing channels) [42, 34, 43]. Under repetition, even tiny per-operation advantages can amplify, motivating explicit accounting (Section 3).

Anonymous-DHT context. Congestion-EQ is complementary to lookup-path anonymity and query privacy in DHT ecosystems (e.g., Octopus and robust-DHT query privacy; PIR-style private routing for IPFS) [11, 12, 13]. Most deployed routing follows Kademlia [14]; recent work studies lightweight obfuscation and hashing-based variants in Kademlia/IPFS ecosystems [15, 16, 17]. Structured P2P anonymity systems built atop DHT-style routing and random-walk delegation (e.g., Salsa and ShadowWalker) and their lookup leakage tradeoffs have also been analyzed [19, 20, 23]. Broader “anonymous lookup” design space (including DHT-based relay discovery pitfalls) is surveyed in classic work [21, 22]. Deployed low-latency anonymity systems also maintain DHT-like directories (e.g., I2P’s netDB); measurement and attack work on those directories further motivates treating congestion and directory-probing load as first-class leakage surfaces [24, 25]. Recent IPFS ecosystem studies also highlight anonymity/abuse pressures tied to provider records and gateway/pinning service surfaces [18].

Contributions.

- We formalize congestion observables as a separate leakage tier and define a TV contract (Congestion-EQ) that composes with the series’ lookup-compiler calculus.
- We derive analytic structure for the M/M/1 waiting-time witness and connect it to slack sensitivity, including evidence that epochs must lengthen (or flushes must be stronger) as slack shrinks.
- We give compositional rules for multi-hop stages and for epoch/repeat accounting, with an “odometer” view designed for certificate binding [5].
- We connect the contract to an Anonymous-DHT microstudy and provide robustness evidence beyond M/M/1 (heavy-tailed service, bursty/clustered arrivals, and binary loss/ECN witnesses including finite-buffer dependence).

2 Model and leakage tier

We assume an overlay lookup compiler that maps a high-level lookup request to a randomized schedule of network actions. Prior papers in this series treat destination/contact traces as the primary channel; here we define a separate congestion channel.

Adjacency. We inherit the series’ adjacency relation on secrets: typically two secrets s, s' differ by one lookup target/key or by one membership bit in a declared neighboring set, and the contract must hold uniformly over that adjacency class [1]. Congestion-EQ does not require a particular adjacency definition, but it does require that the deployment and certificate bind the intended adjacency semantics so that budgets are comparable across releases.

Definition 1 (Congestion observables). Fix an epoch and an observation function producing a random object Z from runtime behavior, e.g., an RTT time series, a queueing-delay sample stream, ECN/loss marks, or summary statistics. Given two adjacent secrets s, s' and any fixed public randomness, the induced laws are $\mathcal{L}(Z \mid s)$ and $\mathcal{L}(Z \mid s')$.

Definition 2 (Congestion-EQ contract (TV form)). A compiler instance satisfies ε -Congestion-EQ for a specified congestion witness family if for all adjacent secrets s, s' ,

$$\text{TV}(\mathcal{L}(Z \mid s), \mathcal{L}(Z \mid s')) \leq \varepsilon \quad (\text{per operation, per epoch}),$$

together with an explicit accounting rule describing (i) how ε composes under repeats, (ii) how it resets under declared flushes, and (iii) how public randomness is exposed to both adversary and verifier.

3 Amplification: small bias, big leak

We use TV because (i) it upper-bounds the bias of *any* (possibly randomized) test, and (ii) it composes cleanly. Concretely, for any measurable test $\varphi(Z) \in [0, 1]$,

$$|\mathbb{E}[\varphi(Z) \mid s] - \mathbb{E}[\varphi(Z) \mid s']| \leq \text{TV}(\mathcal{L}(Z \mid s), \mathcal{L}(Z \mid s')) \leq \varepsilon,$$

so under equal priors the best single-shot success probability is at most $(1 + \varepsilon)/2$ (standard TV characterizations; see, e.g., [44]). Under m independent repetitions (e.g., i.i.d. witnesses), TV can grow quickly: for product laws $P^{\otimes m}, Q^{\otimes m}$ with $\text{TV}(P, Q) \leq \varepsilon$, a coupling argument gives

$$\text{TV}(P^{\otimes m}, Q^{\otimes m}) \leq 1 - (1 - \varepsilon)^m \leq m\varepsilon,$$

so distinguishability scales roughly linearly for small ε and then saturates near 1. This motivates explicit accounting rather than informal “small” claims.

Remark 1 (Dependence and stateful queues). Congestion witnesses are often *not* independent across operations: queue state and congestion control create history dependence. In this regime, the product-law bound above is only a heuristic; Congestion-EQ instead uses the history-conditioned epoch telescoping rule (Proposition 2) to account across repeats without assuming independence.

Figure omitted in source-only release. Requested file: `fig_attack_tv_amplification.pdf`.

Figure 1: Empirical amplification microstudy from the research archive: repeated operations can turn a small per-op advantage into a decisive test. This motivates an explicit accountant rather than informal “small” claims.

Figure omitted in source-only release. Requested file: `fig_mm1_slack_required.pdf`.

Figure 2: Slack sensitivity in M/M/1 from the archive: the slack required to keep TV below a fixed budget rises quickly as utilization approaches 1. (See associated JSON in `../data`.)

4 Slack sensitivity in M/M/1

In a basic M/M/1 abstraction of the congestion witness (queueing delay), the waiting-time law changes sharply with utilization ρ .

Lemma 1 (M/M/1 waiting-time witness). *For an M/M/1 queue with arrival rate λ , service rate μ and $\rho = \lambda/\mu < 1$, the steady-state waiting time W of an arrival satisfies $\mathbb{P}[W = 0] = 1 - \rho$ and, for $t > 0$, $\mathbb{P}[W > t] = \rho e^{-(\mu-\lambda)t}$. Thus W is a point mass at 0 plus an exponential tail with rate $\mu - \lambda = \mu(1 - \rho)$. A small fractional load shift $\lambda \mapsto (1 + \eta)\lambda$ changes the exponential rate by $\Delta(\mu - \lambda) = -\eta\mu\rho$, so the relative change in the tail parameter scales like $\eta\rho/(1 - \rho)$ and blows up as $\rho \rightarrow 1$. See, e.g., standard queueing references [6, 7].*

The archive derives and evaluates closed forms linking slack $(1 - \rho)$ to required “headroom” for a target TV budget, and the figures below summarize that slack sensitivity.

5 Other congestion witnesses: threshold marks and loss/ECN

Delay samples are not the only congestion observable. Many transports expose binary congestion signals (loss, ECN marks, CoDel/PIE-style marking, queue-length thresholds, etc.). Even coarse signals can leak: for M/M/1 in steady state, the queue-length tail satisfies $\mathbb{P}[L \geq T] = \rho^T$. If a public marking rule emits $\text{mark} = \mathbf{1}\{L \geq T\}$, then the induced witness is Bernoulli with parameter $p(\rho) = \rho^T$. For adjacent secrets modeled as a small multiplicative load change $\rho \mapsto (1 + \eta)\rho$, the single-sample TV is simply $|p(\rho) - p((1 + \eta)\rho)|$.

5.1 Finite buffers: loss probability and loss/mark streams

Many real systems have finite buffers, and loss itself is observable. In an M/M/1/K loss system with capacity K (including service), the steady-state occupancy satisfies $p_n \propto \rho^n$ with $\rho = \lambda/\mu$, so the blocking probability is

$$p_{\text{block}}(\rho) = p_K(\rho) = \frac{(1 - \rho)\rho^K}{1 - \rho^{K+1}} \quad (\rho \neq 1), \quad p_K(1) = \frac{1}{K + 1}.$$

See standard queueing texts for M/M/1/K blocking formulas and PASTA (Poisson Arrivals See Time Averages) [8, 7, 9]. By PASTA, an arriving job is blocked with this probability. Thus a one-bit loss witness $B \in \{0, 1\}$ has single-sample TV $\text{TV}(B) = |p_K(\rho_0) - p_K(\rho_1)|$ under adjacent hypotheses. Fig. 5 shows that finite-buffer loss becomes highly sensitive as ρ increases and K grows. Even if a single arrival is only weakly informative, repeated loss/mark samples can amplify advantage, motivating explicit accounting for *loss/ECN mark streams* as a congestion witness family.

Figure omitted in source-only release. Requested file: `fig_queue_forgetting_vs_slack.pdf`.

Figure 3: Queue “forgetting” versus slack: empirical evidence that epochs must lengthen (or flushes must be stronger) as slack shrinks. This underpins reset/flush rules in the accountant.

Figure omitted in source-only release. Requested file: `fig_ecn_threshold.tv.pdf`.

Figure 4: New evidence (added in this split archive): threshold-mark (ECN-like) witness leakage versus utilization in M/M/1. Even a one-bit congestion signal becomes highly sensitive as $\rho \rightarrow 1$, especially for larger thresholds T . This supports treating *loss/ECN/time-series marks* as first-class congestion witnesses alongside delay.

Additional loss/mark evidence (archive; on request). The research archive includes dependence-aware loss/mark witnesses (windowed counts, run/burst statistics, and simple dependence baselines). In this unified source-only archive we summarize the key takeaway in Section 10; full plots and JSON outputs are available on request.

6 Queue-TV stability, composition, and an accountant

Congestion-EQ must support composition across *time* (repeats/epochs) and across *stages* (multi-hop overlays).

A principled way to select epoch length and reset/flush rules is to combine empirical “forgetting” curves (Fig. 3) with conservative stability bounds for the underlying queue-state Markov process. Perturbation theory for uniformly ergodic Markov chains provides TV-distance bounds under small kernel changes (e.g., load drift), making explicit how mixing rate controls how quickly secret-dependent state differences can persist across epochs [10]; see also Levin–Peres–Wilmer for standard coupling/TV mixing background [44]. We use this as a justification for exporting epoch parameters as compiler outputs that can be audited and re-tuned.

In general, if a witness is assembled from multiple components (e.g., per-hop delay streams or end-to-end RTT), then TV admits an additive upper bound that is tight in the worst case.

Proposition 1 (Stage composition (history-conditioned TV)). *Let Z_1, Z_2 be congestion witnesses for two sequential stages (e.g., two hops in a circuit) under adjacent secrets s, s' . Assume $\text{TV}(\mathcal{L}(Z_1 | s), \mathcal{L}(Z_1 | s')) \leq \varepsilon_1$ and, for every realized z_1 ,*

$$\text{TV}(\mathcal{L}(Z_2 | Z_1 = z_1, s), \mathcal{L}(Z_2 | Z_1 = z_1, s')) \leq \varepsilon_2.$$

Then for the concatenated witness $Z_{1:2} = (Z_1, Z_2)$,

$$\text{TV}(\mathcal{L}(Z_{1:2} | s), \mathcal{L}(Z_{1:2} | s')) \leq \varepsilon_1 + \varepsilon_2.$$

Moreover, any measurable function of $Z_{1:2}$ (e.g., an end-to-end RTT/sojourn witness) has TV at most $\varepsilon_1 + \varepsilon_2$ by data processing.

Proof. Write P and Q for the joint laws of (Z_1, Z_2) under s and s' . By a standard telescoping/coupling argument,

$$\text{TV}(P, Q) \leq \text{TV}(P_{Z_1}, Q_{Z_1}) + \sup_{z_1} \text{TV}(P_{Z_2|Z_1=z_1}, Q_{Z_2|Z_1=z_1}),$$

and the assumed bounds give the claim. Data processing gives the final statement. $\square$

Figure omitted in source-only release. Requested file: `fig_mm1k_loss_tv.pdf`.

Figure 5: New evidence (added in this update): finite-buffer loss as a congestion witness. Single-arrival TV between Bernoulli blocking indicators in M/M/1/K under a 1% load shift, for several buffer capacities K . The annotation reports an i.i.d. Binomial amplification approximation at one representative point (useful as a lower-effort sanity-check; real loss streams can be dependent).

Figure omitted in source-only release. Requested file: `fig_dht_destination_by_congestion.pdf`.

Figure 6: Archive microstudy: even when destination/contact traces are controlled, congestion observables can retain path information. This motivates treating congestion as a first-class leakage tier in Anonymous-DHT compilers.

Congestion-EQ is intended as a *contract* between compiler and deployment: the compiler exposes knobs (calendar period, dither, substitution rules, flush strength), while the deployment provisions slack. The research archive contains a compositional rule set and an additive “odometer” accountant for repeats/resets.

Proposition 2 (Epoch/repeat composition (history-conditioned TV)). *Let operations be partitioned into epochs $e = 1, \dots, E$ and let H_{e-1} denote the public transcript and any verifier-visible history up to the start of epoch e . Assume a history-conditioned per-epoch bound: for every history value h ,*

$$\text{TV}(\mathcal{L}(Z_e \mid H_{e-1} = h, s), \mathcal{L}(Z_e \mid H_{e-1} = h, s')) \leq \varepsilon_e.$$

Then the full witness $Z_{1:E}$ satisfies

$$\text{TV}(\mathcal{L}(Z_{1:E} \mid s), \mathcal{L}(Z_{1:E} \mid s')) \leq \sum_{e=1}^E \varepsilon_e.$$

This bound does not require independence across epochs; it is designed for stateful congestion processes where the auditor conditions on published epoch boundaries and public randomness.

Proof. Iterate the telescoping argument over epochs, conditioning on H_{e-1} at each step. □

7 AnonDHT mapping: destination-by-congestion gap

Anonymous DHT lookups can equalize destination/contact traces via delegation, scheduling, and termination-time hiding (series prerequisite). However, congestion witnesses can still correlate with lookup paths, bucket choices, or subtle load differences. This is largely orthogonal to lookup-privacy mechanisms that hide or obfuscate query keys and provider discovery (e.g., Octopus and robust-DHT query privacy), as well as query-obfuscation hardening of Kademlia and PIR-style private content discovery for IPFS [11, 12, 15, 13, 16]. Routing attacks (e.g., routing-table poisoning and eclipse/Sybil-style disruption) can also *induce* secret-correlated load skews in DHT-like ecosystems, making congestion witnesses easier to exploit [26, 27, 28].

8 Limitations and scope

This paper intentionally isolates a TV-tier contract. An adversary may also use other statistics (e.g., spectral/periodicity tests) when calendars are visible; those are treated as additional tiers in the mechanism paper and in the series’ spectral-tier analysis [3].

Figure omitted in source-only release. Requested file: `fig_mm1k_loss_window_tv.pdf`.

Figure 7: Archive evidence: TV between loss-count-in-window laws in an event-driven $M/M/1/K$ simulation (with an i.i.d. Binomial sanity check).

9 Conclusion

Congestion observables are a powerful and distinct metadata channel. Congestion-EQ provides a practical TV-based contract that integrates with compiler calculus work and supports verifiable accounting.

10 Robustness and two-hop microstudies (summary; plots on request)

The main text uses an $M/M/1$ slack-sensitivity calculation as the cleanest analytic “unit test.” In the original split archive, we also included small robustness microstudies (not shipped in this source-only bundle) to check that the phenomenon is not an artifact of memoryless assumptions. We summarize the takeaways here.

Service-time robustness ($M/G/1$). With heavy-tailed service (e.g., lognormal with the same mean but larger variance), the qualitative behavior persists: for a fixed *fractional* load shift (e.g., a 1% increase in effective arrival rate), the induced change in the waiting-time law grows rapidly as slack $1 - \rho$ shrinks. Intuitively, in the Pollaczek–Khinchine picture the waiting-time tail is controlled by moments of the service law and the gap to saturation; when $1 - \rho$ is small, even tiny parameter drift produces a large change in the stationary delay distribution.

Arrival-process robustness (bursty/on-off). With bursty arrivals (two-state on/off or MMPP-style models), slack sensitivity can be *worse*: parallel fanout, clustered retries, and correlated request phases create higher short-term utilization spikes, which amplify delay differences that are invisible to mean-rate models. Operationally, this means a congestion contract should be validated under *burst* models, not only under Poisson baselines.

Two-hop (stage) composition. We stated an additive stage bound for end-to-end TVs. Microstudies on two-hop pipelines (including clustered-burst arrivals) were consistent with the bound: end-to-end TV stayed below the per-hop TV sum, and the bound became closer to tight when both hops operated near the slack “knee” and perturbations were aligned.

Loss/mark witnesses (finite buffers): dependence and window statistics. In the split archive, we also evaluated one-bit congestion witnesses such as loss and ECN marks under finite buffers (e.g., $M/M/1/K$). A key practical point is that these streams are *dependent*: once the buffer is full, consecutive drops can occur, and burst/run statistics become a witness on their own. Windowed summaries (e.g., loss-count-in-window and max-run-length-in-window) can amplify faster than i.i.d. Bernoulli sanity checks suggest. Operationally, if loss/mark streams are within scope, budget them explicitly and publish monitoring summaries that include dependence-sensitive diagnostics (not only mean loss).

Figure omitted in source-only release. Requested file: `fig_mm1k_loss_runlength_tv.pdf`.

Figure 8: Archive evidence: dependence-sensitive loss witness (max consecutive-loss run length in a window) can deviate from i.i.d. baselines.

What to do with this. Treat slack as a *first-class knob*: if you want a small congestion witness, you need explicit slack margin and a declared load envelope. The omitted plots, scripts, and raw tables that support these bullets are available on request (and were previously shipped as supplements in the split congestion archive).

11 Artifacts and reproducibility

This unified archive is intentionally *source-only*: it ships a single-file L^AT_EX manuscript with all external figures replaced by boxed placeholders. The plots, scripts, and raw tables used for the microstudies referenced in this paper (including material that previously lived in split-archive supplements) are available on request as an external artifact bundle. The referee-facing claim is the TV-tier contract and its accounting rules; the artifact bundle exists to support implementer validation and to supply regression vectors for auditors.

12 Takeaways

- **Congestion is a witness:** RTT/queueing delay/loss/ECN marks can carry metadata even when payloads are encrypted; treat them as explicit transcript projections with budgets.
- **Congestion-EQ exports a budget/accountant:** a TV-style distance on congestion traces plus an accountant for composition across repeated queries and windows.
- **Mechanisms live elsewhere:** use PSC-Q for compiler-friendly mechanisms that meet Congestion-EQ budgets; use W-Congestion-EQ for proof-carrying certificates and transparency anchoring.
- **Non-redundancy:** later papers should cite Congestion-EQ for witness definitions and budgets rather than re-defining “delay leakage” locally.

References

- [1] Anonymous. *Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups*. Draft v0.67 (2026-01-21). Series paper. Published. Wiki: `[[2026.01.22 - Mathematics: Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups]]`.
- [2] Anonymous. *Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy*. January 2026. Published. Wiki: `[[2026.01.29 - Mathematics: Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy]]`.
- [3] Anonymous. *Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays*. January 2026. Series paper. Published. Wiki: `[[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]]`.

- [4] Anonymous. *PSC-Q: Public Service Calendars and Slack-Harvesting Substitution Padding for Congestion-Private Anonymous Overlays*. Companion paper in this unified archive.
- [5] Anonymous. *W-Congestion-EQ: Proof-Carrying Congestion Budgets with Certificates and Transparency Logs for Anonymous Overlays*. Companion paper in this unified archive.
- [6] D. G. Kendall. Stochastic processes occurring in the theory of queues and their analysis by the method of the imbedded Markov chain. *The Annals of Mathematical Statistics*, 24(3), 1953.
- [7] S. Asmussen. *Applied Probability and Queues*. Springer, 2nd edition, 2003.
- [8] D. Gross, J. F. Shortle, J. M. Thompson, and C. M. Harris. *Fundamentals of Queueing Theory*. Wiley, 4th edition, 2008.
- [9] R. W. Wolff. Poisson arrivals see time averages. *Operations Research*, 30(2):223–231, 1982. DOI: 10.1287/opre.30.2.223.
- [10] A. Yu. Mitrophanov. Sensitivity and convergence of uniformly ergodic Markov chains. *Journal of Applied Probability*, 42(4):1003–1014, 2005. https://alexmitr.com/Mitrophanov_Uniformly_Ergodic_MCs_JAP_2005.pdf.
- [11] Q. Wang and N. Borisov. Octopus: A secure and anonymous DHT lookup. In *ICDCS*, 2012. DOI: 10.1109/ICDCS.2012.78. <https://nikita.ca/papers/octopus-icdcs12.pdf>.
- [12] M. Backes, I. Goldberg, A. Kate, and T. Toft. Adding query privacy to robust DHTs. In *ASIACCS*, 2012, pp. 30–31. DOI: 10.1145/2414456.2414473. <https://arxiv.org/abs/1107.1072>.
- [13] M. Mazmudar, S. Veitch, and R. Akhavan Mahdavi. Peer2PIR: Private Queries for IPFS. In *IEEE Symposium on Security and Privacy (S&P)*, 2025. DOI: 10.1109/SP61157.2025.00231. <https://arxiv.org/abs/2405.17307>.
- [14] P. Maymounkov and D. Mazières. Kademlia: A peer-to-peer information system based on the XOR metric. In *International Workshop on Peer-to-Peer Systems (IPTPS)*, 2002. DOI: 10.1007/3-540-45748-8_5. <https://pdos.csail.mit.edu/~petar/papers/maymounkov-kademlia-lncs.pdf>.
- [15] E. Daniel, G. Michel, and F. Tschorsch. Improving Kademlia Lookup Privacy through Query Obfuscation. In *ACM/SIGAPP Symposium on Applied Computing (SAC)*, 2025, pp. 1345–1352. DOI: 10.1145/3672608.3707703. <https://dl.acm.org/doi/pdf/10.1145/3672608.3707703>. <https://fis.tu-dresden.de/portal/en/publications/improving-kademlia-lookup-privacy-through-query-obfuscation%2834f33c16-7610-4c43-9ba1-028fd028f033%29.html>.
- [16] T. Katsantas, Y. Thomas, C. Karapapas, and G. Xylomenos. Enhancing IPFS privacy through triple hashing. In *IEEE Symposium on Computers and Communications (ISCC)*, 2024. DOI: 10.1109/ISCC61673.2024.10733729. <https://mm.aueb.gr/publications/5b9cdb00-fa7d-4e93-b2d0-31f509116371.pdf>.
- [17] M. Wedler, E. Daniel, and F. Tschorsch. Plausibly Deniable Content Discovery for Bitswap Using Random Walks. In *IEEE 49th Conference on Local Computer Networks (LCN)*, 2024. DOI: 10.1109/LCN60385.2024.10639726. <https://arxiv.org/abs/2407.12433>.

- [18] C. Karapapas, I. Pittaras, G. C. Polyzos, and C. Patsakis. Hello, won't you tell me your name?: Investigating Anonymity Abuse in IPFS. arXiv:2506.04307, 2025. <https://arxiv.org/abs/2506.04307>.
- [19] A. Nambiar and M. Wright. Salsa: A Structured Approach to Large-Scale Anonymity. In *ACM CCS*, 2006, pp. 17–26. DOI: 10.1145/1180405.1180409. <https://www.freehaven.net/anonbib/cache/Salsa.pdf>.
- [20] P. Mittal and N. Borisov. ShadowWalker: Peer-to-Peer Anonymous Communication Using Redundant Structured Topologies. In *ACM CCS*, 2009. DOI: 10.1145/1653662.1653683.
- [21] R. Morselli. *Lookup Protocols and Techniques for Anonymity*. PhD dissertation, University of Maryland, 2006. <https://www.cs.umd.edu/~jkatz/THESES/ruggero.pdf>.
- [22] Q. Wang, P. Mittal, and N. Borisov. In Search of an Anonymous and Secure Lookup: Attacks on Structured Peer-to-Peer Anonymous Communication Systems. In *ACM CCS*, 2010. DOI: 10.1145/1866307.1866343. <https://www.freehaven.net/anonbib/cache/ccs10-lookup.pdf>.
- [23] P. Mittal and N. Borisov. Information Leaks in Structured Peer-to-Peer Anonymous Communication Systems. *ACM Transactions on Information and System Security*, 15(1):5:1–5:28, 2012. DOI: 10.1145/2133375.2133380.
- [24] C. Egger, J. Schlumberger, C. Kruegel, and G. Vigna. Practical Attacks Against the I2P Network. In *RAID*, 2013. https://sites.cs.ucsb.edu/~chris/research/doc/raid13_i2p.pdf.
- [25] J. P. Timpanaro, T. Cholez, I. Chrisment, and O. Festor. Evaluation of the Anonymous I2P Network's Design Choices Against Performance and Security. In *ICISSP*, 2015. <https://www.scitepress.org/papers/2015/52266/52266.pdf>.
- [26] B. Prünster, A. Marsalek, and T. Zefferer. Total Eclipse of the Heart – Disrupting the InterPlanetary File System. In *31st USENIX Security Symposium (USENIX Security)*, 2022, pp. 3735–3752. <https://www.usenix.org/system/files/sec22-prunster.pdf>.
- [27] A. Singh, T.-W. Ngan, P. Druschel, and D. S. Wallach. Eclipse Attacks on Overlay Networks: Threats and Defenses. In *IEEE INFOCOM*, 2006. DOI: 10.1109/INFOCOM.2006.231. <https://people.mpi-sws.org/~druschel/publications/Eclipse-Infocom.pdf>.
- [28] L. Wang and J. Kangasharju. Real-World Sybil Attacks in BitTorrent Mainline DHT. In *IEEE Global Communications Conference (GLOBECOM)*, 2012. DOI: 10.1109/GLOCOM.2012.6503215. <https://nymity.ch/sybilhunting/pdf/Wang2012a.pdf>.
- [29] S. Gast, M. Lipp, L. Brümmer, M. Göbel, S. Mangard, and D. Gruss. SnailLoad: Exploiting Remote Network Latency Measurements without Direct Observation of Traffic. In *USENIX Security Symposium*, 2024. <https://www.usenix.org/system/files/usenixsecurity24-gast.pdf>.
- [30] S. J. Murdoch and G. Danezis. Low-Cost Traffic Analysis of Tor. In *IEEE Symposium on Security and Privacy*, 2005. DOI: 10.1109/SP.2005.12. <https://murdoch.is/papers/oakland05torta.pdf>.

- [31] S. Gast, N. Puntigam, S. Franza, S. R. Neela, D. Gruss, and J. Ullrich. Zero-Click SnailLoad: From Minimal to No User Interaction. In *ESORICS*, 2025. DOI: 10.1007/978-3-032-07901-5_6. <https://gruss.cc/files/zeroclicksnailload.pdf>.
- [32] N. S. Evans, R. Dingledine, and C. Grothoff. A practical congestion attack on Tor using long paths. In *USENIX Security Symposium*, 2009. <https://freehaven.net/anonbib/papers/congestion-longpaths.pdf>.
- [33] N. Hopper, E. Y. Vasserman, and E. Chan-Tin. How much anonymity does network latency leak? In *ACM CCS*, 2007. DOI: 10.1145/1315245.1315257. <https://www.freehaven.net/anonbib/cache/ccs07-latency-leak.pdf>.
- [34] X. Gong, N. Kiyavash, and N. Borisov. Quantifying information leakage in timing side channels. arXiv:1403.1276, 2014.
- [35] M. AlSabah, K. Bauer, I. Goldberg, D. Grunwald, D. McCoy, S. Savage, and G. M. Voelker. DefenestraTor: Throwing Out Windows in Tor. In *PETS*, 2011. DOI: 10.1007/978-3-642-22263-4_8.
- [36] Tor Project. Proposal 324: RTT Congestion Control. Tor design proposal, 2020–. <https://spec.torproject.org/proposals/324-rtt-congestion-control.html>.
- [37] A. Mishra, L. Rastogi, R. Joshi, and B. Leong. Keeping an Eye on Congestion Control in the Wild with Nebby. In *ACM SIGCOMM*, 2024. DOI: 10.1145/3651890.3672223. <https://cs.stanford.edu/~keithw/sigcomm2024/sigcomm24-final110-acmpaginated.pdf>.
- [38] R. Ware, A. A. Philip, N. Hungria, Y. Kothari, J. Sherry, and S. Seshan. CCAalyzer: An Efficient and Nearly-Passive Congestion Control Classifier. In *ACM SIGCOMM*, 2024. DOI: 10.1145/3651890.3672255. <https://www.justinesherry.com/papers/ware-sigcomm2024.pdf>.
- [39] W. Wang, D. Xue, P. Kumar, A. Mishra, and R. Ensafi. Is Custom Congestion Control a Bad Idea for Circumvention Tools? *Free and Open Communications on the Internet (FOCI)*, 2025. <https://www.petsymposium.org/foci/2025/foci-2025-0001.pdf>.
- [40] A. Sabzi, R. Vora, S. Goswami, M. Seltzer, M. Lécuyer, and A. Mehta. NetShaper: A Differentially Private Network Side-Channel Mitigation System. In *USENIX Security Symposium*, 2024. <https://www.usenix.org/system/files/sec24summer-prepub-168-sabzi.pdf>.
- [41] A. Mehta, H. N. Nguyen, S. Friedman, and collaborators. Pacer: Comprehensive Network Side-Channel Mitigation in the Cloud. In *USENIX Security Symposium*, 2022. Available from USENIX.
- [42] V. Anantharam and S. Verdú. Bits through queues. *IEEE Transactions on Information Theory*, 42(1):4–18, 1996.
- [43] A. Beams, S. Kannan, and S. Angel. Packet Scheduling with Optional Client Privacy. In *ACM CCS*, 2021, pp. 3415–3430. DOI: 10.1145/3460120.3485371. <https://www.cis.upenn.edu/~sga001/papers/ifs-ccs21.pdf>.
- [44] D. A. Levin and Y. Peres, with contributions by E. L. Wilmer. *Markov Chains and Mixing Times*, second edition. American Mathematical Society, 2017. <https://pages.uoregon.edu/dlevin/MARKOV/markovmixing.pdf>.

- [45] R. Jansen. CellShift: RTT-Aware Trace Transduction for Real-World Website Fingerprinting. In *NDSS*, 2026. DOI: 10.14722/ndss.2026.231004. <https://www.ndss-symposium.org/ndss-paper/cellshift-rtt-aware-trace-transduction-for-real-world-website-fingerprinting/>.
- [46] Anonymous. Artifact policy (source-only archive; artifacts available on request). Series note, 2026.
- [47] Anonymous. Notation and Minimal Model for the One-True-Archive. Series synthesis note (Synthesis 8), 2026. (This archive.)